25
15
175
357
525

Page .....
Date .....

# Chapter 8

1) Multiply in Base 6

$(35)_6 \times (15)_6$

35
x13
303
350
$(453)_6$

First Partial product
$5 \times 3 = 11_6$ → Carry 1
$5 \times 3 + 1 = 21_6$ ⇒ Carry 3
First product = 311
Second ⇒ $35 \times 1$
So answer $(1101)_6$

Multiplication in base 6 was harder because its number system is less familiar than base 10 and carrying is done at 6 instead of 10

2) The video demystifies microprocessors by providing that transistors are not magical electronic devices — they are simply basic switches. Whether a switch is powered by microscopic electrical voltage or a human raising a physical flag combining millions of on / off switches together according to simple logic rules allows a machine to calculate anything